Satwik Paladugula

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GitHub | LinkedIn

Profile

Computer Science undergraduate with strong foundational expertise in **Java, SQL, Object-Oriented Programming, and Database Management Systems**, coupled with hands-on experience building **AI-powered applications and scalable backend microservices**. Certified in ServiceNow (CAD/CSA) and SAP ABAP Cloud, with demonstrated experience developing production-grade solutions using Java, Python, Spring Boot, FastAPI, and MySQL. Passionate about software engineering, intelligent systems, and continuous learning.

Education

B.Tech — Computer Science and Technology 2023–2027
Sri Vasavi Engineering College, Tadepalligudem CGPA: 8.25/10

Intermediate (PCM) 2021–2023
Bharatiya Vidya Bhavans 84.8%

SSC 2020–2021
Tirumala Proactive School 10/10 CGPA

Certifications

- **ServiceNow Certified Application Developer (CAD)**
- **ServiceNow Certified System Administrator (CSA)**
- SAP Certified - Back-End Developer - ABAP Cloud
- Oracle Generative AI Professional
- Salesforce Agentforce Specialist

Technical Skills

- **Programming Languages:** Java, SQL, Python, C
- **Backend & Frameworks:** Spring Boot, Spring Data JPA, Hibernate, FastAPI, RESTful APIs
- **Databases & Storage:** MySQL, SQLite, DBMS, Relational Database Design
- **Developer Tools:** Git, GitHub, Postman, VS Code, Linux
- **Core Subjects:** OOP, DBMS, Data Structures, Operating Systems, Computer Networks
- **Professional Skills:** Problem Solving, Excellent Verbal and Written Communication, Teamwork

Projects

AI Security Gateway — Enterprise LLM Proxy & Prompt Defense Python, FastAPI, SQLite

- Built a high-performance reverse proxy between enterprise clients and LLMs to prevent confidential data leakage and defend against prompt injection attacks.
- Implemented algorithmic PII detection (Luhn checksum, Shannon entropy) adding only 0.14 ms p50 latency, with multi-turn token masking and 90% unit test coverage.

CollectFast — 0% Fee UPI Invoicing & WhatsApp Recovery Engine Python, FastAPI, ReportLab

- Architected an automated B2B invoicing SaaS enabling businesses to create GST tax invoices and collect direct payments with 0% gateway fees via native NPCI UPI protocol.
- Implemented dynamic real-time UPI QR generation and mobile deep-linking triggers (upi://pay?pa=...) for instant 1-tap checkout via Google Pay, PhonePe, and Paytm.
- Built a 1-click WhatsApp payment recovery engine with automated behavioral reminder templates, accelerating client payment recovery times by 70%.

ClarifySQL — AI-Powered Natural Language Data Assistant Python, FastAPI, Gemini, MySQL

- Developed an AI-powered data assistant that converts natural language business questions into secure, schema-aware SQL queries for MySQL databases.
- Implemented query understanding, ambiguity detection, interactive clarification workflows, and strict AST-level SQL validation using Python and SQLGlot.
- Built a FastAPI backend and Streamlit interface featuring dynamic query result visualization, persistent execution history, and session-based conversational context.

Student Management System Java, Spring Boot, MySQL

- Developed a RESTful backend application using Spring Boot and MySQL for efficient student record management.
- Implemented CRUD operations using Spring Data JPA and Hibernate for reliable database interaction.
- Tested REST APIs using Postman and managed source code using Git and GitHub.

Strengths

- **Problem Solving:** Strong analytical and logical thinking skills
- **Quick Learner:** Ability to quickly learn new technologies, tools, and enterprise platforms
- **Teamwork & Communication:** Effective collaboration with clear verbal and written communication